
CS246: Database Management Systems Lab

Lab # 12 (1 Questions, 100 Marks)

Lab session: AL1

Held on: 05-Apr-2024 (Sat)

Lab Timings: 14:00 to 17:00 Hours Pages: 4

Submission time: 16:45 Hrs, 05-Apr-2025

Instructors Dr. V. Vijaya Saradhi

Head TAs Adithya K Moorthy & Laxita Agrawal

Department of CSE, IIT Guwahati

1. This lab theme is database replication
2. Read the supplementary material provided

Question 1: (100 points)

Perform the following tasks.

Task 01a (1 mark) On master server: create database **week12a** and **week12b**

Task 01b (1 mark) On slave server: create database **week12a** and **week12b**

Task 02 (10 marks) **Master Server Configuration:** Perform the steps described in the word document. Important: for task 01 to task 10 use the database **week12a**.

Task 03 (20 marks) **Slave Server Configuration:** Perform the steps described in the word document.

Task 04a (3 marks) On master server:

1. (1 marks) The **sailor** table has the following description

column #	name	data type	constraint
1	sid	int	primary key
2	sname	char(50)	None
3	rating	int	None
4	age	decimal(3,1)	None

2. (1 marks) The **boat** table has the following description.

column #	name	data type	constraint
1	bid	int	primary key
2	bname	char(50)	None
3	bcolor	char(50)	None

3. (1 marks) The **reserve** table has the following description.

column #	name	data type	constraint
1	sid	int	refers to sailor
2	bid	int	refers to boat
3	day	date	None
			sid, bid, day is primary key
			sid foreign key cascade on update
			sid foreign key cascade on delete
			bid foreign key cascade on update
			bid foreign key cascade on delete

Task 04b (3 marks) On slave server: Demonstrate through SQL query the replicated tables

Task 05a (3 marks) On master server:

1. Insert data from **sailors.csv** into **sailor** table
2. Insert data from **boats.csv** into **boat** table
3. Insert data from **reserves.csv** into **reserve** table

Task 05b (3 marks) On slave server: Demonstrate through SQL query the replicated data in the respective tables.

Task 06a (6 marks) On master server:

1. (1 mark) Alter table **sailor** to add one column **earning int** after **sname** with default value 100.
2. (1 mark) Alter table **boat** to add one column **make char(20)** after **bcolor** with default value **Toyota**.
3. (1 mark) Alter table **reserve** to add one column **amount int** after **day** with default value 10.
4. (1 mark) Insert one row with random data into **sailor** table
5. (1 mark) Insert one row with random data into **boat** table
6. (1 mark) Insert one row with random data into **reserve** table

Task 06b (3 marks) On slave server: Demonstrate through SQL query the replicated data in the respective tables.

Task 07a (3 marks) On master server:

1. (1 mark) Delete **earning** column from **sailor** table
2. (1 mark) Delete **make** column from **boat** table
3. (1 mark) Delete **amount** column from **reserve** table

Task 07b (3 marks) On slave server: Demonstrate through SQL query the replicated data in the respective tables.

Task 08a (4 marks) On master server:

1. Write a stored procedure `insert_sailors` which takes no arguments and inserts 10000 random records of sailors into `sailor` table

Task 08b (4 marks) On slave server: Demonstrate display the slave status through `SHOW SLAVE STATUS \G`. Observe the fields

Task 09a (6 marks) On master server:

1. (4 marks) Write a trigger on `boat` table and after insert, this trigger should insert at least 10 records in `reserve` table for the newly inserted boat with random `sid` and random `day`
2. (2 marks) Insert two new boats with random `bid`, `bname`, `bcolor` into `boat` table

Task 09b (2 marks) On slave server: Demonstrate the effect trigger on `reserve` table.

Task 10a (3 marks) On master server: Create a view `sailor_reserve_boat` containing columns `sname`, `bname`, `day`

Task 10b (1 mark) On slave server: Show the view.

Task 11a (10 marks) **Replication Filtering:** This task involves sharing only one table of `week12b` database.

Master Server: Perform this task only after completing task 01 to task 10. Do not attempt task 11 as your first task. For task 11 use the database `week12b`.

1. (3 marks) Create three tables `sailor`, `boat`, `reserve` identical to Task 03a description and populate data from files `sailors.csv`, `boats.csv`, `reservers.csv`
2. (3 marks) Create a table `reserve_details` having the following description

column #	name	data type	constraint
1	<code>sname</code>	<code>char(50)</code>	refers to <code>sailor</code>
2	<code>bname</code>	<code>char(50)</code>	refers to <code>boat</code>
3	<code>day</code>	<code>date</code>	None
<code>sname</code> , <code>bname</code> , <code>day</code> is primary key <code>sname</code> foreign key cascade on update <code>sname</code> foreign key cascade on delete <code>bname</code> foreign key cascade on update <code>bname</code> foreign key cascade on delete			

3. (3 marks) Edit master server configuration file
`binlog_do_db = week12b`
4. (1 mark) Restart master server.

Task 11b (5 marks) On slave server:

1. Create a table `reserve_details` identical to Task 11a, point 2.
2. (2 marks) **Slave Server configuration:** Perform the following: Edit the file `/etc/mysql/mysql.conf.d/mysqld.cnf` using root privileges as follows

```
# Replication filters
binlog-do-db = week12b
replicate-do-table = week12b.reserve_details
```

3. (1 mark) Restart Slave server.
4. (6 marks) Refer to the document `database-replication.pdf`, perform step 3 and step 4 of Table Level Filtering (page 7).
5. (1 mark) Show the contents of `reserve_details`
6. Show the contents of `reserve_details` after updating `sailor` table on master server
7. (1 mark) Show the contents of `reserve_details` after deletion of a record in `sailor` table on master server
8. (1 mark) Show the contents of `reserve_details` after updating `boat` table on master server
9. (1 mark) Show the contents of `reserve_details` after deletion of a record in `boat` table on master server

Task 12 (3 marks) Stop replication on slave.

Task 13 (3 marks) Stop MySQL sever on Master and Slave.